

Chapter

2

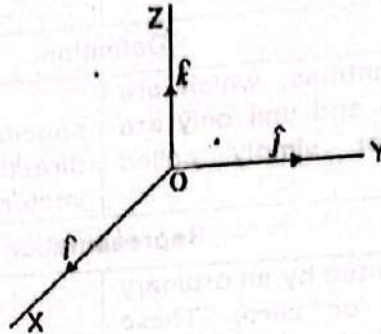
Basics of Scalar and Vectors

SCALARS	VECTORS
Definition	
Those Physical quantities, which are specified by their magnitude and unit only are called scalar quantities or simply called SCALARS .	Those Physical quantities which are specified by their magnitude and as well as direction with appropriate units are called vector quantities or simply called vector.
Representation	
Scalars are represented by an ordinary number (positive, negative or zero). These numbers are known as magnitude of scalars. The magnitude is also known as value.	The vector is represented by a letter with an Arrow like $\vec{A}$ or $\vec{B}$ etc.
Denotion	
They are denoted by letters in ordinary type. They do not require any mention of direction for their specification and representation.	They are denoted by bold faced letters with an arrow, i.e. $\vec{A}$ and $\vec{B}$ their magnitudes are denoted by $ \vec{A} $ or A and $ \vec{B} $ or B respectively.
Required Method	
Scalars are added, subtracted, multiplied and divided by ordinary arithmetical rules.	Vector operations such as addition, subtraction, and multiplication do not follow basic arithmetic rules; instead, they require vector algebra for accurate computation.
Example	
Temperature, length, speed, time density, mass etc. are the examples of scalars.	Displacement, velocity, acceleration, force, moment of force, etc.

REPRESENTATION OF VECTORS:

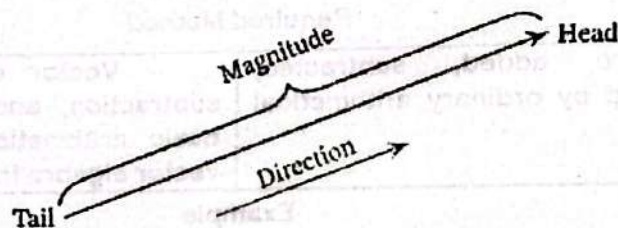
- **Symbolic (Analytical) Representation:**
The symbolic representation of vectors often uses variables and symbols to represent direction and magnitude. For example:
- **Magnitude:** A vector $\vec{A}$ with magnitude A can be symbolically represented as: $|\vec{A}|$ or A .
- **Direction:** A vector $\vec{B}$ with a specific direction can be represented symbolically with an arrow Or Bold Face Letter **B**

- **Components:** Components of a vector $\vec{A}$ along the x, y, and z axes can be represented symbolically as $\vec{A}_x$, $\vec{A}_y$ and $\vec{A}_z$, respectively.
- **Unit Vectors:** Unit vectors in the direction of the coordinate axes can be represented as $\hat{i}$, $\hat{j}$, and $\hat{k}$ for the x, y, and z directions, respectively also Known as Rectangular unit Vector.



- **Graphical Representation:**

Graphical representation of vectors involves the use of arrows and line segments to visually depict vectors in both magnitude and direction.



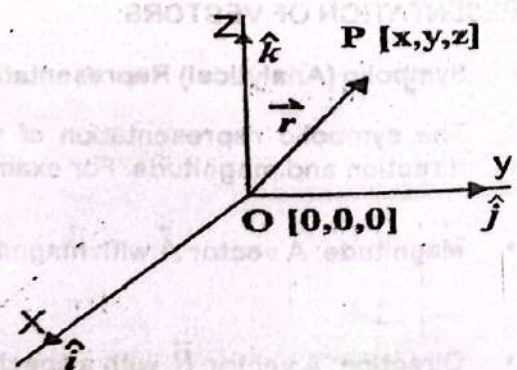
TYPES OF VECTORS:

Position Vector:

A vector, which determines the position of a point relative to the fixed point (Origin) is called position vector.

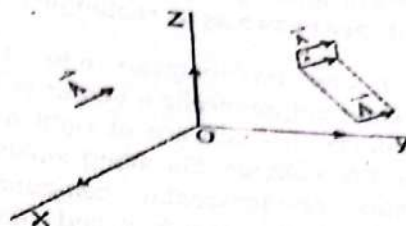
Vector". Consider a fixed reference point 'O' and specify the position of a given point 'P' with respect

to the point 'O' by means of vector having magnitude and direction represented by a directed line as $\vec{OP}$. This can be shown by the following figure. A position vector is given as: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$



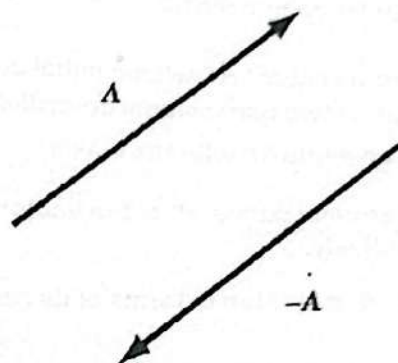
Free Vector:

A vector which is independent of origin and can be displaced parallel to itself & applied at any point is known as Free vector

**Negative Of A Vector:**

The negative of a vector is a vector with the same magnitude but opposite direction to the original vector.

Example: If $\vec{A}$ is a vector the $-\vec{A}$ will have same magnitude but opposite direction.

**Null Vector:**

If two vectors are identical in magnitude and opposite in direction, then the difference vector is called Null or ZERO vectors"

The null vector has zero magnitude and has no direction or it may have any direction.

For e.g : The displacement of a footballer is 15m due east then it comes back to its initial point by covering the same displacement but in opposite direction.

As

$$\vec{A} + (-\vec{A}) = 0 \text{ (Null Vector)}$$

Chapter 2

Unit Vector and Resolution of Vector

UNIT VECTOR:

A vector, whose magnitude is unity, i.e. ($A = 1$) in any given direction is called unit vector.

FORMULA

Consider a vector ' $\vec{A}$ ' whose unit vector is represented by ' $\hat{a}$ '. The unit vector ' $\hat{a}$ ' can be obtained by dividing the vector ' $\vec{A}$ ' by its magnitude i.e.

$$\hat{a} = \frac{\vec{A}}{A}$$

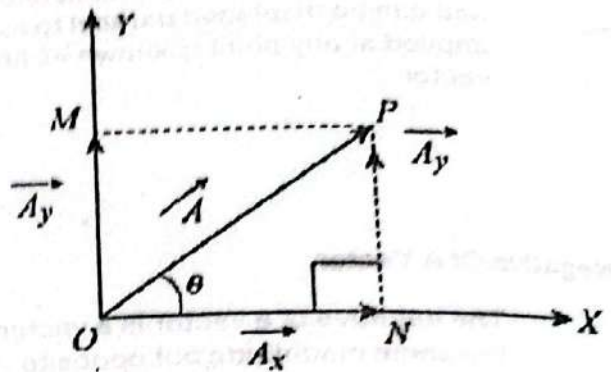
Physics XI

RESOLUTION OF VECTOR:

The process of splitting a vector into its components is known as resolution of vector and the components are known as its rectangular component.

A vector can be resolved into a number of components, but generally a vector is resolved into two components which are at right angle to each other i.e. the components along x-axis is called x-component or horizontal components & the components along y-axis is called y-component or vertical components. Such components are called rectangular components.

Consider a vector ' $\vec{A}$ ' whose initial point is placed at the origin of two dimensional co-ordinate system and making an angle ' θ ' with the x-axis



From the terminal points 'P' of the vector draw two perpendiculars lines $\overline{PN}$ and $\overline{PM}$ on x-axis and y-axis respectively.

The vector ' $\vec{A}$ ' is written in terms of its rectangular components as

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

Magnitude Of Rectangular Components

By using the trigonometric ratios, the magnitudes of horizontal & vertical components can be obtained, i.e.

Horizontal Component: From $\triangle ONP$: $\cos \theta = \frac{ON}{OP}$ or $ON = OP \cos \theta$

$$A_x = A \cos \theta \text{ or } |\vec{A}_x| = A \cos \theta$$

Vertical Component: From $\triangle ONP$: $\sin \theta = \frac{PN}{OP}$ or $PN = OP \sin \theta$

$$A_y = A \sin \theta \text{ or } |\vec{A}_y| = A \sin \theta$$

Magnitude Of The Resultant Vector

From the Pythagoras theorem, we can easily get the magnitude of resultant vector ' $\vec{A}$ ', i.e.

$$\text{From } \triangle ONP: (OP)^2 = (ON)^2 + (PN)^2$$

$$|\vec{A}| = \sqrt{|\vec{A}_x|^2 + |\vec{A}_y|^2} \text{ or } A = \sqrt{A_x^2 + A_y^2}$$

Direction

The Direction of ' $\vec{A}$ ' is determined by θ which can be found out by the following formula

$$\tan \theta = \frac{|\vec{A}_y|}{|\vec{A}_x|} \text{ or } \theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$

Chapter

2

Addition of Vectors

ADDITION OF VECTORS:

"The method of combining two or more vectors to get a single vector is called addition of the vectors."

Classification

Addition of vectors can be classified into two ways

Addition Of Two Parallel/Opposite Vectors

- If vectors are parallel then the magnitude are simply added.
- If vectors are anti parallel then the magnitude are subtracted. The direction of the resultant is along the vector of larger magnitude.

Addition Of Non Parallel Vectors

When vectors are at some angle then the addition of vectors can be classified into two ways.

- Graphical Method or Head to tail rule (For more than two vectors)
- By Law of parallelogram method. (For only two vectors)
- By Rectangular component method. (For more than two vectors)

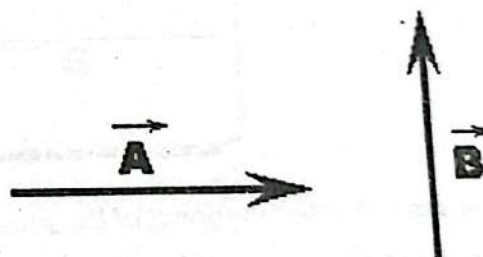
ADDITION OF VECTORS BY GRAPHICAL METHOD OR HEAD TO TAIL RULE:

Consider two vectors $\vec{A}$ and $\vec{B}$ acting in the directions as shown below

In order to get their resultant vector by head to tail method we must follow the following steps

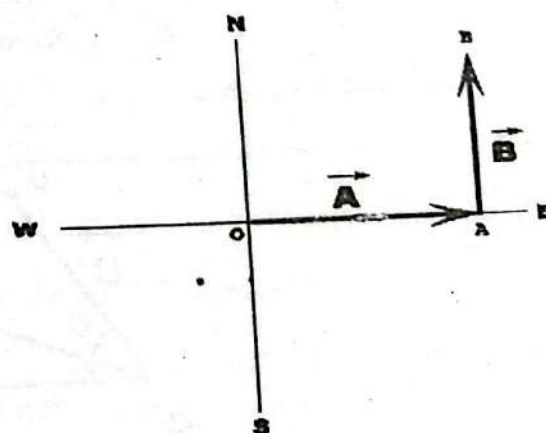
STEP # 1

Choose a suitable scale for the vectors so that they can be plotted.

STEP # 2

Draw representative line OA of vector $\vec{A}$.

Draw representative line AB of vector $\vec{B}$ such that the tail of $\vec{B}$ coincides with the head of vector $\vec{A}$.

STEP # 3

Join 'O' and 'B'.
OB represents resultant vector of given vectors $\vec{A}$ and $\vec{B}$

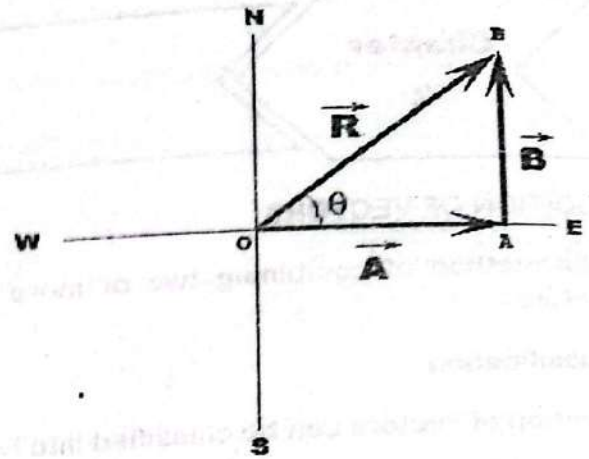
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STEP #4

Measure the length of line segment OB and multiply it with the scale chosen initially to get the magnitude of resultant vector.

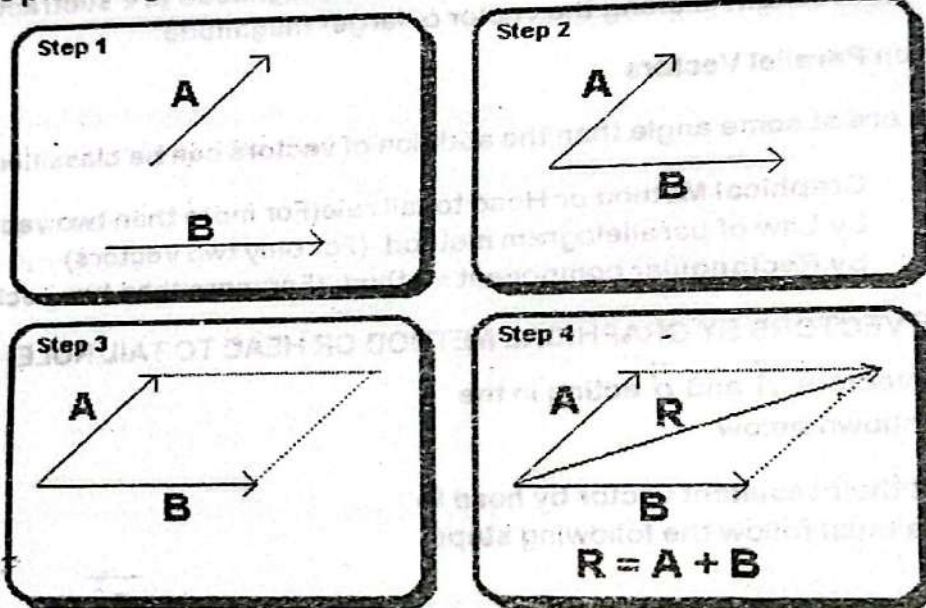
STEP #5

The direction of the resultant vector is directed from the tail of vector $\vec{A}$ to the head of vector $\vec{B}$.



ADDITION OF VECTORS BY PARALLELOGRAM METHOD:

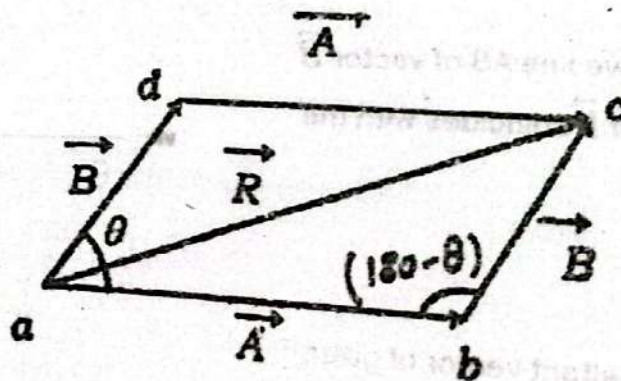
The parallelogram method for vector addition, the vectors are translated, (i.e., moved) to a common origin and the parallelogram constructed as follows:



The resultant R is the diagonal of the parallelogram drawn from the common origin,

Magnitude of Resultant: The magnitude of resultant vector " $\vec{R}$ " can be determined by the law of cosine according to which

$$R = \sqrt{A^2 + B^2 - 2AB \cos \angle \text{Opp to } R}$$

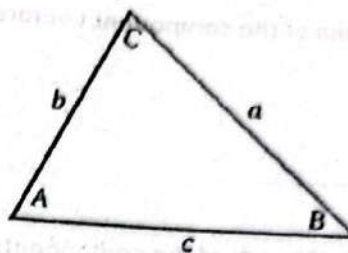


In triangle $\triangle ABC$ the angle opposite to $\vec{R}$ is $(180 - \theta)$ so $\cos(180 - \theta) = -\cos \theta$

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

The direction of resultant can be determined by the law of sine's, which states that :

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$



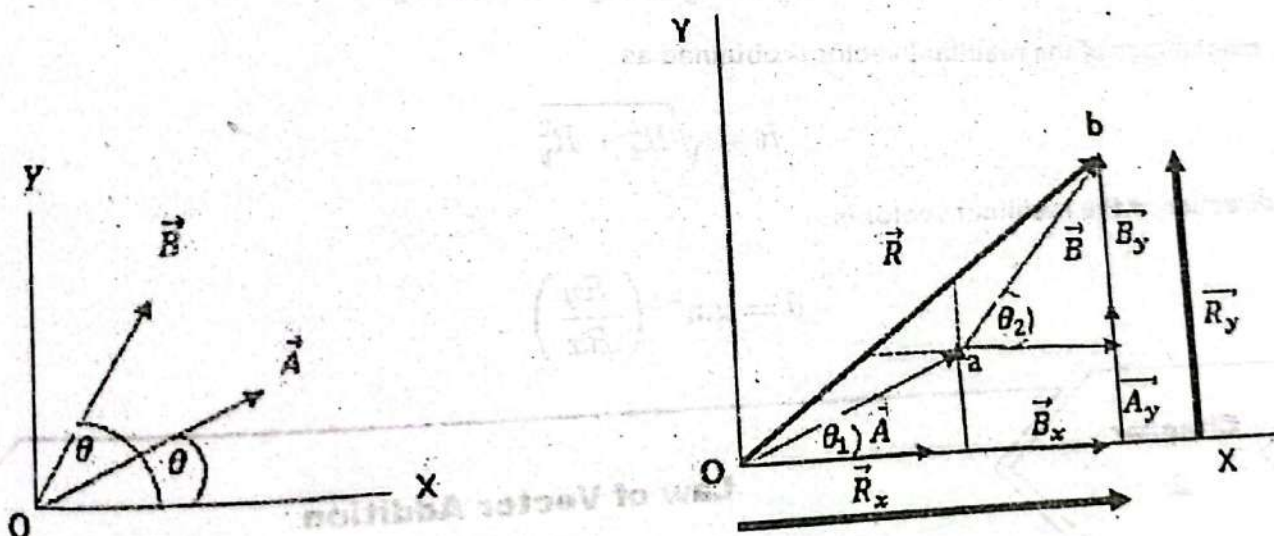
Chapter 2

Addition of Vectors By Rectangular Components

ADDITION OF VECTOR BY RECTANGULAR COMPONENT:

Consider two vectors $\vec{A}$ and $\vec{B}$ which are making angles ' θ_1 ' and ' θ_2 ' with the positive x-axis.

when $\vec{A}$ and $\vec{B}$ are added by head-to-tail rule, then we obtain the resultant vector $\vec{R}$.



Now resolve the vector $\vec{A}$ into its components $\vec{A}_x$ and $\vec{A}_y$. The magnitudes of these components are as follows.

$$A_x = A \cos \theta_1$$

$$A_y = A \sin \theta_1$$

Similarly the vector $\vec{B}_x$ and $\vec{B}_y$ & their magnitudes are as follows:

$$B_x = B \cos \theta_2$$

$$B_y = B \sin \theta_2$$

The sum of the component vectors along x-axis is equal to the x-components of resultant vector. $\vec{R}$ i.e.

$$\vec{R}_x = A_x \hat{i} + B_x \hat{i}$$

$$\vec{R}_x = (A_x + B_x) \hat{i}$$

Similarly the sum of the component vectors along y-axis is

$$\vec{R}_y = A_y \hat{j} + B_y \hat{j}$$

$$\vec{R}_y = (A_y + B_y) \hat{j}$$

Now the sum of the magnitudes of x-components is equal to the magnitude of the x-component of resultant vector, i.e.

$$R_x = A_x + B_x$$

$$R_x = A_x \cos \theta_1 + B_x \cos \theta_2$$

Similarly the sum of the magnitudes of y-components is

$$R_y = A_y + B_y$$

$$R_y = A_y \sin \theta_1 + B_y \sin \theta_2$$

The magnitudes of the resultant vector is obtained as

$$R = \sqrt{R_x^2 + R_y^2}$$

The direction of the resultant vector is

$$\theta = \tan^{-1} \left(\frac{R_y}{R_x} \right)$$

Chapter 2

Law of Vector Addition

COMMUTATIVE LAW OF VECTOR ADDITION:

Consider a parallelogram. OACB Let the two vectors $\vec{A}$ and $\vec{B}$ represent the two adjacent sides of the parallelogram. The diagonal $\vec{OC}$ of the parallelogram represents the resultant vector.

From figure $\triangle OQP$ and $\triangle OQS$

$$\vec{A} + \vec{B} = \vec{R}$$

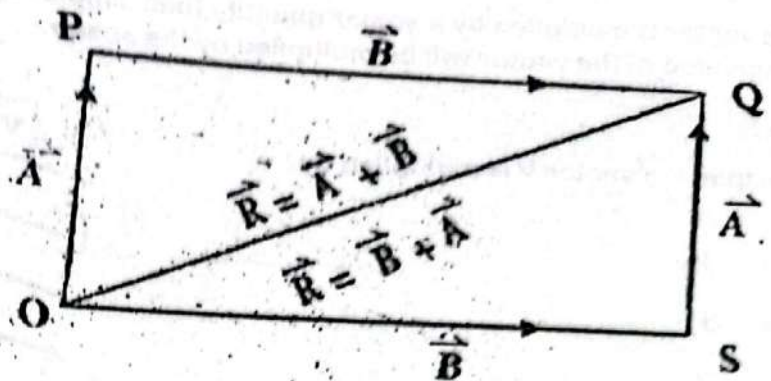
&

$$\vec{B} + \vec{A} = \vec{R}$$

Therefore

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

This is known as Commutative Law.



ASSOCIATIVE LAW OF VECTOR ADDITION

Consider the following figure in which OP represents the vector $\vec{A}$, PQ represents vector $\vec{B}$, QS represents the vector $\vec{C}$ and OS represents the vector $\vec{R}$,

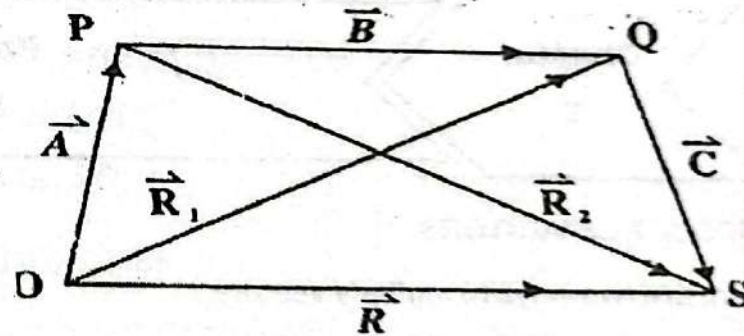
From the $\triangle OQS$, in which $\vec{OQ}$ represents the resultant $\vec{A} + \vec{B}$, which is obtained by using "head to tail" rule.

Thus

$$(\vec{A} + \vec{B}) + \vec{C} = \vec{R}$$

Similarly in $\triangle OPS$, the line $\vec{PS}$ represents the resultant as

$$\vec{B} + \vec{C}$$



$$\vec{A} + (\vec{B} + \vec{C}) = \vec{R}$$

Therefore

$$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$$

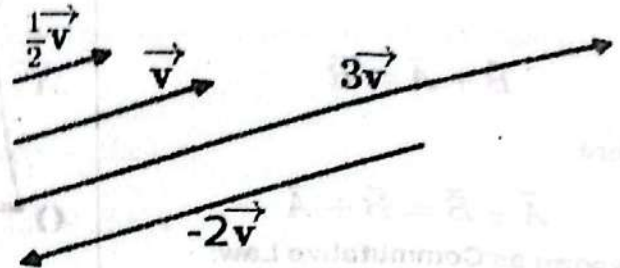
This is known as associative law of vector addition.

MULTIPLICATION OF A VECTOR BY SCALAR:

If a vector is multiplied by a scalar quantity then simple algebraic rules are applied and the magnitude of the vector will be multiplied by the scalar.

Example: If a vector $\vec{V}$ is multiplied by

- $\frac{1}{2}$
- 3
- -2



Rules to remember:

If a vector $\vec{A}$ is multiplied by m then it will obey

1. Commutative law:
2. Associative law
3. Distributive law

$$m \vec{A} = \vec{A} m$$

$$m(n \vec{A}) = (m n) \vec{A}$$

$$(m + n) \vec{A} = m \vec{A} + n \vec{A} \text{ or } m(\vec{A} + \vec{B}) = m \vec{A} + m \vec{B}$$

Chapter 2

Product of Vectors Scalar Product

PRODUCT OF VECTORS

There are two ways to multiply vectors

- Scalar Product
- Vector Product

SCALAR PRODUCT OR DOT PRODUCT

"If two vector are multiplied & their product is a scalar, then the product is called scalar product
Dot Product"

OR

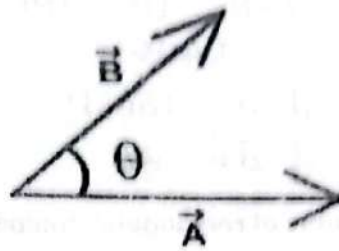
"The product of magnitudes of two vectors and the cosine of the angle between them is called scalar product or Dot Product"

Mathematical Expression

Consider two vector $\vec{A}$ & $\vec{B}$ having angle ' θ ' between them, then their dot product is mathematically expressed as

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

Where the quantity ' $AB \cos \theta$ ' is a scalar, therefore this product is called scalar product and is also called dot product of two vectors $\vec{A}$ and $\vec{B}$.



EXAMPLES

Work is the dot product of force and displacement so work itself a scalar quantity

$$W = \vec{F} \cdot \vec{d}$$

Power is the dot product of force and velocity so power itself a scalar quantity

$$P = \vec{F} \cdot \vec{V}$$

CHARACTERISTICS OF DOT PRODUCT

(i) If vector $\vec{A}$ and $\vec{B}$ are parallel i.e. $\theta = 0$, then

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\vec{A} \cdot \vec{B} = AB \cos(0)$$

$$\vec{A} \cdot \vec{B} = AB(1)$$

$$\vec{A} \cdot \vec{B} = AB$$

(ii) If $\vec{A} = \vec{B}$, i.e. $\vec{A}$ is parallel and equal to $\vec{B}$, ($\theta = 0^\circ$), then

If there is two similar vectors i.e their magnitude and their direction are same then ,

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\vec{A} \cdot \vec{A} = AA \cos(0)$$

$$\vec{A} \cdot \vec{A} = A^2(1)$$

$$\vec{A} \cdot \vec{A} = A^2$$

(iii) If $\vec{A}$ & $\vec{B}$ are perpendicular to each other i.e. $\theta = 90^\circ$ or any one of the two vectors is zero.

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

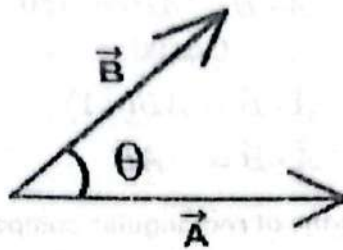
$$\vec{A} \cdot \vec{B} = AB \cos 90$$

$$\because \cos 90 = 0$$

$$\vec{A} \cdot \vec{B} = AB(0)$$

$$\vec{A} \cdot \vec{B} = 0$$

Where the quantity ' $AB \cos \theta$ ' is a scalar, therefore this product is called scalar product and is also called dot product of two vectors $\vec{A}$ and $\vec{B}$.



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$$\vec{A} \cdot \vec{A} = AA \cos(0)$$

$$\vec{A} \cdot \vec{A} = A^2(1)$$

$$\vec{A} \cdot \vec{A} = A^2$$

(iii) If $\vec{A}$ & $\vec{B}$ are perpendicular to each other i.e. $\theta = 90^\circ$ or any one of the two vectors is zero.

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\vec{A} \cdot \vec{B} = AB \cos 90$$

$$\because \cos 90 = 0$$

$$\vec{A} \cdot \vec{B} = AB(0)$$

$$\vec{A} \cdot \vec{B} = 0$$

(iv) If $\vec{A}$ & $\vec{B}$ are antiparallel to each other i.e. $\theta = 180^\circ$ then product is negative.

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\vec{A} \cdot \vec{B} = AB \cos 180$$

$$\therefore \cos 180 = -1$$

$$\vec{A} \cdot \vec{B} = AB(-1)$$

$$\vec{A} \cdot \vec{B} = -AB$$

v) The dot product of two vectors in terms of rectangular components

$$\text{If } \vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \text{ and } \vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

then,

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

vi) If the unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ are perpendicular and parallel to each other, then

$$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{i} \cdot \hat{k} = 0$$

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$$

Chapter

2

Product of Vectors Vector Product

VECTOR PRODUCT OR CROSS PRODUCT:

"If two vectors are multiplied and their product is a vector quantity then the product is called vector product or cross product".

Mathematical Expression

Consider two vector $\vec{A}$ and $\vec{B}$ and the product of these two vectors is denoted by $\vec{A} \times \vec{B}$, that's why it is read as $\vec{A}$ cross $\vec{B}$ and the product of these two vectors gives a new vector $\vec{C}$. Mathematically it can be expressed as

$$\vec{A} \times \vec{B} = \vec{C}$$

Magnitude Of Vector

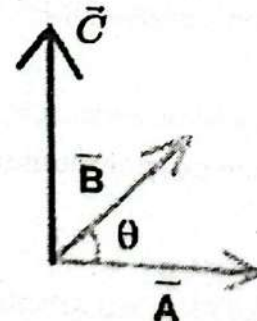
The magnitude of vector $\vec{C}$ is given by

$$|\vec{A} \times \vec{B}| = AB \sin \theta = |\vec{C}|$$

or

$$C = AB \sin \theta$$

Where ' θ ' is the smaller angle between positive direction of $\vec{A}$ and $\vec{B}$.



(iv) If $\vec{A}$ & $\vec{B}$ are antiparallel to each other i.e. $\theta = 180^\circ$ then product is negative.

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\vec{A} \cdot \vec{B} = AB \cos 180$$

$$\therefore \cos 180 = -1$$

$$\vec{A} \cdot \vec{B} = AB(-1)$$

$$\vec{A} \cdot \vec{B} = -AB$$

v) The dot product of two vectors in terms of rectangular components

$$\text{If } \vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \text{ and } \vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

then,

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

vi) If the unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ are perpendicular and parallel to each other, then

$$\hat{i} \cdot \hat{j} = \hat{k} \cdot \hat{j} = \hat{i} \cdot \hat{k} = 0$$

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$$

Chapter

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Product of Vectors

Vector Product

VECTOR PRODUCT OR CROSS PRODUCT:

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Mathematical Expression

Consider two vector $\vec{A}$ and $\vec{B}$ and the product of these two vectors is denoted by $\vec{A} \times \vec{B}$, that's why it is read as $\vec{A}$ cross $\vec{B}$ and the product of these two vectors gives a new vectors $\vec{C}$. Mathematically it can be expressed as

$$\vec{A} \times \vec{B} = \vec{C}$$

Magnitude Of Vector

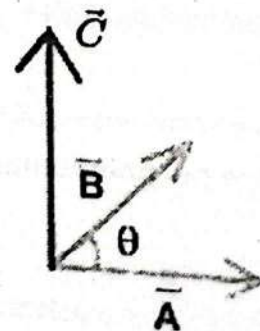
The magnitude of vector $\vec{C}$ is given by

$$|\vec{A} \times \vec{B}| = AB \sin \theta = |\vec{C}|$$

or

$$C = AB \sin \theta$$

Where ' θ ' is the smaller angle between positive direction of $\vec{A}$ and $\vec{B}$.



Direction of Vector:

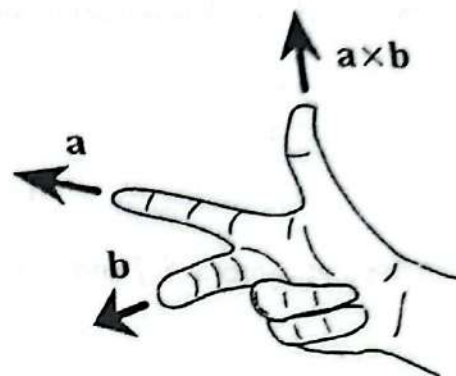
The direction of the resultant vector of a vector product (cross product) depends on the right-hand rule. The right-hand rule is a convention used to determine the direction of the product vector in a cross product operation. Here's how it works:

Imagine holding your right hand in a specific orientation while working with vectors:

Extend your right hand INDEX finger in the direction of the first vector (A).

Extend your right MIDDLE finger in the direction of the second vector (B).

Now, the direction in which your right THUMB points represents the direction of the vector $\vec{C}$ or $\vec{A} \times \vec{B}$

**Example Of Vector Product**

- Torque is the cross product of force and moment arm ($\vec{\tau} = \vec{r} \times \vec{F}$)
- Angular momentum is cross product of linear momentum and position vector.

$$\vec{l} = \vec{r} \times \vec{p}$$

$$\vec{l} = \vec{r} \times m\vec{v}$$

Characteristics of Vector Product:

- (i) Vector Product does not obey commutative law

$$\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A} \text{ Or } \vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$$

- (ii) Vector Product is distributive over addition

$$\vec{A} \times (\vec{B} + \vec{C}) = (\vec{A} \times \vec{B}) + (\vec{A} \times \vec{C})$$

- (iii) If vector $\vec{A}$ and $\vec{B}$ are parallel i.e. $\theta = 0^\circ$, or antiparallel i.e. $\theta = 180^\circ$ then their product is zero

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{u}$$

$$\vec{A} \times \vec{B} = AB \sin(0) \hat{u}$$

$$\vec{A} \times \vec{B} = 0$$

Similarly, if

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{u}$$

$$\vec{A} \times \vec{B} = AB \sin(180) \hat{u}$$

$$\vec{A} \times \vec{B} = 0$$

- (iv) If unit vectors are parallel to each other.

$$\hat{i} \times \hat{i} = 0$$

$$\hat{j} \times \hat{j} = 0$$

$$\hat{k} \times \hat{k} = 0$$

- (v) If $\vec{A} = \vec{B}$, i.e. $\vec{A}$ is perpendicular to $\vec{B}$ ($\theta = 90^\circ$), then their product is maximum.

$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

$$\vec{A} \times \vec{B} = AB \sin 90^\circ \hat{n}$$

$$\vec{A} \times \vec{B} = AB(1)\hat{n} = AB\hat{n}$$

- (v) If the unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ mutually perpendicular to each other.

$$\hat{i} \times \hat{j} = \hat{k} \quad \text{or} \quad \hat{j} \times \hat{i} = -\hat{k}$$

$$\hat{j} \times \hat{k} = \hat{i} \quad \text{or} \quad \hat{k} \times \hat{j} = -\hat{i}$$

$$\hat{k} \times \hat{i} = \hat{j} \quad \text{or} \quad \hat{i} \times \hat{k} = -\hat{j}$$

- (vii) If $\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$ and $\vec{B} = B_1\hat{i} + B_2\hat{j} + B_3\hat{k}$ are written in terms of their rectangular components. Then the cross product of $\vec{A}$ and $\vec{B}$ is

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} A_2 & A_3 \\ B_2 & B_3 \end{vmatrix} - \hat{j} \begin{vmatrix} A_1 & A_3 \\ B_1 & B_3 \end{vmatrix} + \hat{k} \begin{vmatrix} A_1 & A_2 \\ B_1 & B_2 \end{vmatrix}$$

$$= (A_2B_3 - A_3B_2)\hat{i} - (A_1B_3 - A_3B_1)\hat{j} + (A_1B_2 - A_2B_1)\hat{k}$$

Chapter

2

Speed, Velocity and Acceleration

SPEED:

Definition: Speed is a scalar quantity that represents the rate at which an object covers a distance. It measures how fast an object is moving without considering its direction.

Symbol: Speed is typically represented by the symbol 'v.'

VELOCITY:

Definition: Velocity is a vector quantity that represents the rate of change of an object's displacement with respect to time. It includes both the speed and the direction of motion.

Symbol: Velocity is typically represented by the symbol ' $\vec{v}$.'

Formula: $v = d / t$.

Average, Instantaneous, and Uniform Aspects:

Average Speed: It is the total distance traveled divided by the total time taken.

$$V_{avg} = \text{Total Distance} / \text{Total Time}$$

Instantaneous Speed: It is the speed of an object at any moment of time. It can vary throughout the motion

$$V_{ins} = \Delta d / \Delta t$$

Where Δt approaching to 0

Uniform Speed: When an object covers equal distances in equal intervals of time, it is said to be moving at a uniform speed.

ACCELERATION:

Definition: Acceleration is a vector quantity that measures how velocity changes with respect to time. It indicates both the magnitude and direction changes with time.

Symbol: Acceleration is typically represented by the symbol ' $\vec{a}$ '.

Formula:

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$$

Types of Acceleration:

- **Uniform Acceleration:**
Uniform acceleration occurs when an object's velocity changes by a constant amount in each unit of time. In other words, the rate of change in velocity remains constant.
- **Non-Uniform Acceleration:**
Non-uniform acceleration occurs when an object's velocity changes at a varying rate over time. The acceleration may be different at different moments.

Average and Instantaneous Acceleration:

- **Average Acceleration:**

Average acceleration measures the overall rate of change in velocity of an object over a specific time interval. It represents the average behavior during that interval. For two dimensional motion.

$$\vec{a}_{avg} = \frac{\Delta v_x}{\Delta t} \hat{i} + \frac{\Delta v_y}{\Delta t} \hat{j} = a_{avg,x} \hat{i} + a_{avg,y} \hat{j}$$

Formula: $\vec{v} = x / t$.

Average, Instantaneous, and Uniform Aspects:

Average Velocity: It is the total displacement divided by the total time taken. It includes both the magnitude and direction of motion.

$$\vec{v}_{avg} = \text{Total Displacement} / \text{Total Time}$$

Instantaneous Velocity: It is the velocity of an object at a specific moment in time, including both magnitude and direction. It can vary throughout the motion

$$\vec{v}_{ins} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$$

Uniform Velocity: When an object covers equal displacements in equal intervals of time in a consistent direction, it is said to be moving at a uniform velocity.

Formula: $v = d / t$.

Average, Instantaneous, and Uniform Aspects:

Average Speed: It is the total distance traveled divided by the total time taken.

$$V_{avg} = \text{Total Distance} / \text{Total Time}$$

Instantaneous Speed: It is the speed of an object at any moment of time. It can vary throughout the motion

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$$\vec{a}_{avg} \equiv \frac{\Delta \vec{v}}{\Delta t}$$

$$\vec{a}_{avg} = \frac{\Delta v_x}{\Delta t} \hat{i} + \frac{\Delta v_y}{\Delta t} \hat{j} = a_{avg,x} \hat{i} + a_{avg,y} \hat{j}$$

Formula: $\vec{v} = x / t$.

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$$\vec{v}_{ins} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}$$

Uniform Velocity: When an object covers equal displacements in equal intervals of time in a consistent direction, it is said to be moving at a uniform velocity.

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- Instantaneous Acceleration:**
Instantaneous acceleration is the acceleration of an object at a specific instant in time. It can vary at different moments and represents the object's acceleration at that particular point. For two dimensional motion.

$$\vec{a} \equiv \lim_{t \rightarrow 0} \vec{a}_{avg} = \lim_{t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

$$\text{Or } \vec{a} = \frac{d\vec{v}}{dt}$$

$$\text{Or } \vec{a} = \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j}$$

$$\text{Or } \vec{a} = a_x \hat{i} + a_y \hat{j}$$

Chapter 2

Equations of Motion

EQUATION OF MOTIONS: (FOR UNIFORM ACCELERATION)

The relationship between displacement, velocity and acceleration are known as equation of motion. These equations are:

$$V_f = V_i + at$$

$$V_f^2 = V_i^2 + 2as$$

$$S = V_i t + \frac{1}{2} at^2$$

Where a is acceleration t is time, S is distance covered, V_i and V_f are initial and final velocities respectively.

For the free fall motion, the equation of motion becomes,

$$V_f = V_i + gt$$

$$V_f^2 = V_i^2 + 2gh$$

$$h = V_i t + \frac{1}{2} gt^2$$

Where g is acceleration due to gravity, t is time, h is height, V_i and V_f are initial and final velocities respectively.

Derivation of First Equation of Motion: ($V = U + at$)

Let us consider $V - T$ graph for a body moving under uniform acceleration and as a result of the acceleration, its velocity increases from u to v in time t . The slope of the $V - T$ graph gives the acceleration of the moving object.

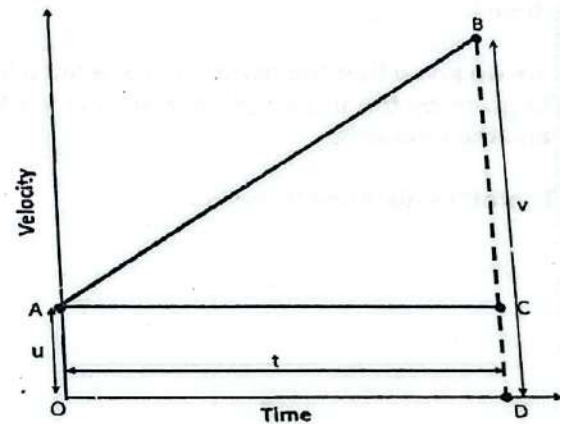
Therefore, Acceleration = slope of $AB = \frac{BC}{CA}$

$$= \frac{BD - CD}{CA} \text{ (as } BC = BD - CD \text{)}$$

$$a = \frac{v - u}{t}$$

$$v - u = at$$

$$v = u + at$$



Derivation of Second Equation of Motion: ($S = ut + \frac{1}{2}at^2$)

Let us consider $V - T$ graph for a body moving under uniform acceleration and as a result of the acceleration, its velocity increases from u to v in time t .

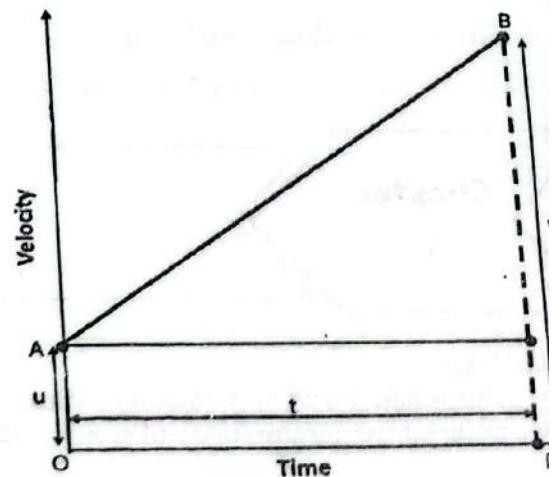
As we know that the distance travelled s in time t is given by the area enclosed by the $v - t$ graph and the time axis.

Therefore distance travelled

$$\begin{aligned} S &= \text{area of trapezium } ABDO \\ &= \text{area of rectangle } ACDO + \text{area of triangle } ABC \\ &= (OA \times OD) + \frac{1}{2} \times AC \times BC \\ &= ut + \frac{1}{2} \times t \times (v - u), [\text{from eq : } v - u = at] \\ &= ut + \frac{1}{2} \times t \times at \end{aligned}$$

Or

$$S = ut + \frac{1}{2}at^2$$

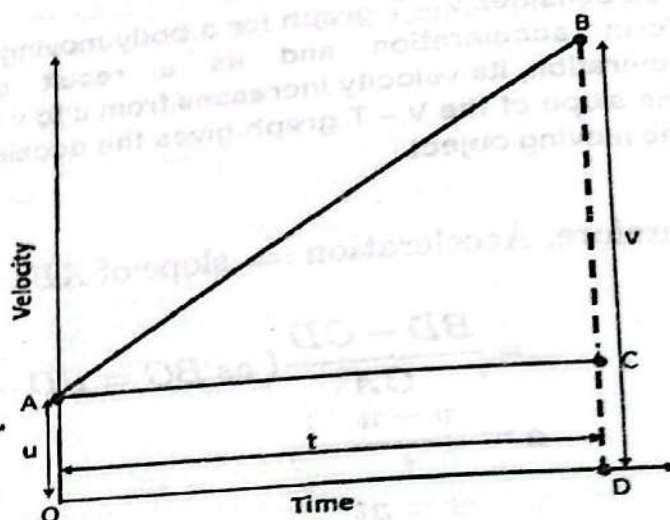


Derivation of Third Equation of Motion: ($v^2 - u^2 = 2as$)

Let us consider $V - T$ graph for a body moving under uniform acceleration and as a result of the acceleration, its velocity increases from u to v in time t .

As we know that the distance travelled s in time t is given by the area enclosed by the $v - t$ graph and the time axis.

Therefore distance travelled



$$s = \text{area of trapezium ABDO}$$

$$= \frac{1}{2} \times (\text{sum of the parallel sides}) \times \text{height}$$

$$= \frac{1}{2} \times (OA + BD) \times OD$$

$$= \frac{(u + v)}{2} \times t$$

$$= \frac{1}{2} \times (v + u) \times \frac{(v - u)}{a} \left[\text{from eq, } t = \frac{(v - u)}{a} \right]$$

$$2as = v^2 - u^2$$

$$v^2 = u^2 + 2as$$

Chapter

2

Projectile Motion

PROJECTILE

"Any object that is given any initial velocity & which subsequently follows a parabolic path determined by the force of gravity acting on it is called a Projectile".

PROJECTILE MOTION

"When a body is thrown with an angle ' θ ' & it covers a (distance) parabolic path under the action of gravity, this type of motion is called Project Motion".

Example

- Kicked or thrown balls
- Jumping animals
- Object thrown from a window
- Object released from an aero plane

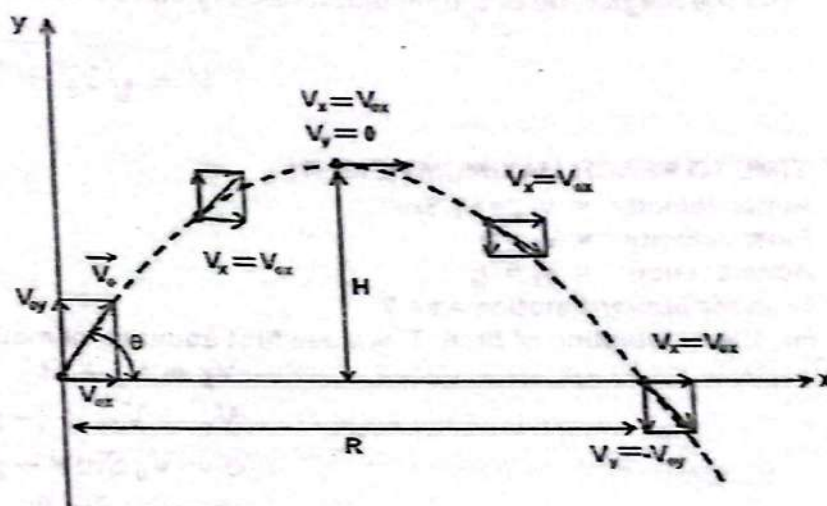
Basic Assumptions Of Projectile Motion

- The value of acceleration due to gravity (g) remains constant throughout the motion.
- Air friction during projectile motion is assumed to be zero.
- There is no effect of earth's rotation on projectile motion.

Derivation:

Suppose a particle is projected at an angle ' θ ' with the horizontal, as shown in figure.

The initial velocity $\vec{V}_0$ of the particle can be resolved into two rectangular components $\vec{V}_{0x}$ & $\vec{V}_{0y}$ along horizontal axis & vertical axis respectively. The magnitudes of the horizontal & vertical components of velocity are as follows.



HORIZONTAL VELOCITY COMPONENT

The horizontal component of initial velocity is given by

$$V_{0x} = V_0 \cos \theta$$

But during the projectile motion, there is no net force acts in the horizontal direction, therefore the final velocity $\vec{V}_x$ in the horizontal direction is equal to its initial velocity $\vec{V}_{0x}$
i.e.

$$V_x = V_{0x} = V_0 \cos \theta$$

VERTICAL VELOCITY COMPONENT

The vertical component of the initial velocity is

$$V_{0y} = V_0 \sin \theta$$

But as the net force acts in vertical direction, which produces acceleration in the y-direction, therefore the final velocity in vertical direction can be calculated with the help of the following data.

Initial Velocity = $V_{0y} = V_0 \sin \theta$

Acceleration = $a_y = -g$

Time = t

Final velocity = $\vec{V}_y = ?$

Using the first equation of motion, i.e.

$$\begin{aligned}
 V_f &= V_i + at \\
 V_y &= V_{oy} + a_y t \\
 V_y &= V_o \sin \theta + (-g)t \\
 V_y &= V_o \sin \theta - gt
 \end{aligned}$$

and the magnitude of the resultant velocity can be calculated by the following formula

$$V = \sqrt{V_x^2 + V_y^2}$$

TIME TO REACH MAXIMUM HEIGHT:

Initial velocity = $V_{oy} = V_o \sin \theta$

Final velocity = $V_y = 0$

Acceleration = $a_y = -g$

Time for upward motion = $t = ?$

For the calculation of time 'T' we use first equation of motion

$$V_f = V_i + at$$

$$V_y = V_{oy} + (-g)t$$

$$0 = V_o \sin \theta - gt$$

$$gt = V_o \sin \theta$$

$$t = \frac{V_o \sin \theta}{g}$$

Where 't' is half of the total time elapsed between launching & landing of the projectile.

TOTAL TIME OF FLIGHT

Total time of flight "T" is doubled the time to reach maximum height.

$$T = 2t$$

$$T = 2 \cdot \frac{V_o \sin \theta}{g}$$

$$T = \frac{2 \cdot V_o \sin \theta}{g}$$

$$T = \frac{2 \cdot V_{oy}}{g}$$

Or

MAXIMUM HEIGHT OF THE PROJECTILE

To derive the maximum height of the projectile first we have to calculate the time for upward motion. The maximum height occurs when the vertical component of the final velocity reduces to zero & the particle is projected with the acceleration due to gravity (-g). Therefore

Initial velocity = $V_{oy} = V_o \sin \theta$

Final velocity = $V_y = 0$

Acceleration = $a_y = -g$

Time for upward motion = $T = ?$

Maximum height = $S = h = ?$

To calculate the maximum height we use the third equation of motion i.e.

$$\begin{aligned}
 2aS &= V_f^2 - V_i^2 \\
 2gh &= V_f^2 - V_i^2 \\
 2gh &= V_y^2 - V_{oy}^2 \\
 2(-g)h &= 0 - (V_o \sin \theta)^2 \\
 -2gh &= -(V_o \sin \theta)^2 \\
 2gh &= (V_o \sin \theta)^2 \\
 h &= \frac{1}{2g} V_o^2 \sin^2 \theta \\
 h &= \frac{1}{2g} V_{Oy}^2
 \end{aligned}$$

Or

RANGE OF THE PROJECTILE

"The horizontal distance from the origin to the point where the projectile returns is called range of the projectile".

It is denoted by 'R'.

In order to find the range of the projectile, we make use of the fact that the total flight takes the time that is twice the time to reach the highest point. Therefore

Distance = S = X = R = ?

Velocity = V = V_{ox}

Using S = V × t

$$\begin{aligned}
 X &= V_{ox} \times T \\
 R &= V_o \cos \theta \times \frac{2 \cdot V_o \sin \theta}{g} \\
 R &= \frac{2 V_o^2}{g} \sin \theta \cos \theta \\
 R &= \frac{V_o^2}{g} 2 \sin \theta \cos \theta
 \end{aligned}$$

But from trigonometry, we know that

$$2 \sin \theta \cos \theta = \sin 2\theta$$

Therefore the above equation becomes

$$R = \frac{V_o^2}{g} \sin 2\theta$$

Thus the range of the projectile depends on the square of the initial velocity & sine of double the projectile angle.

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THE MAXIMUM RANGE

The range is said to be maximum, i.e. $R_{\max}$, when the factor $\sin 2\theta$ in Range equation should be taken maximum, i.e.

$$\sin 2\theta = 1$$

$$2\theta = \sin^{-1}(1)$$

$$2\theta = 90$$

$$\theta = \frac{90}{2}$$

$$\theta = 45^\circ$$

Putting the value of θ in the equation of range we get

$$= \frac{V_0^2}{g} \sin 2(45)$$

$$= \frac{V_0^2}{g} \sin 90^\circ$$

$$= \frac{V_0^2}{g} (1)$$

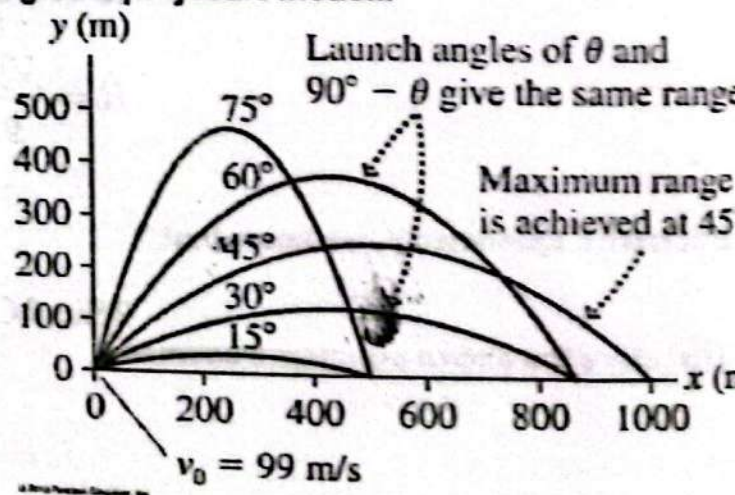
$$R_{\max} = \frac{V_0^2}{g}$$

IMPORTANT NOTE:

The importance of complementary angles ($\theta_2 = 90^\circ - \theta_1$) in projectile motion is that it allows us to launch a projectile with the same range for different angles of projection. This can be useful in a variety of applications, such as sports, military, and engineering.

Here are some other examples of complementary angles in projectile motion:

- 20° and 70°
- 45° and 45°
- 60° and 30°



Chapter

2

Multiple Choice Questions

MCQ'S		ANS.
1. To get a resultant displacement of 10m, two displacement vectors of magnitude 6m and 8m should be combined: (a) Parallel (b) Antiparallel (c) At an angle of 45° (d) Perpendicular to each other		d
2. The velocity of a particle at an instant is 10 m/s and after 5sec the velocity of particle is 20m/s. The velocity 3 sec before in m/s is: (a) 8 (b) 4 (c) 6 (d) 7		b
3. A ball is thrown upwards with a velocity of 100 m/s. It will reach the ground after: (a) 10 sec (b) 20 sec (c) 5 sec (d) 40 sec		b
4. Two projectiles are fired from the same point with the same speed at angles of projection 60° and 30° respectively. Which one of the following is true? (a) The range will be same (b) Their maximum height will be same (c) Their landing velocity will be same (d) Their time of flight will be same		a
5. The ratio of numerical values of average velocity and average speed of a body is always: (a) Unity (b) Unity or less (c) Unity or more (d) Less than unity		a
6. If the average velocities of a body become equal to the instantaneous velocity, body is said to be moving with: (a) Uniform acceleration (b) Uniform velocity (c) Variable velocity (d) Variable acceleration		a
7. At the top of a trajectory of a projectile, the acceleration is: (a) maximum (b) minimum (c) zero (d) g		d
8. At what angle the range of projectile becomes equal to the height of projectile? (a) 65° (b) 45° (c) 76° (d) 30°		c
9. The angle at which dot product becomes equal to the cross product is: (a) 65° (b) 45° (c) 76° (d) 30°		b

10. If the dot product of two non-zero vectors vanishes; the vectors will be:
 (a) in the same direction (b) opposite direction to each other
 (c) perpendicular to each other (d) zero

Exams Practice MCQs

1. If two forces of the same magnitude F are acting at an angle of 120° with each other the magnitude of their resultant will be
 (a) $2F$ (b) 0
 (c) $0.5 F$ (d) F
2. Maximum height of projection depends upon
 (a) Angle of projection. (b) Velocity of projection.
 (c) Both angle and velocity. (d) Range.
3. Which of the following equations represents the relationship between velocity, acceleration, and time in uniformly accelerated motion?
 (a) $v = u + at$ (b) $v = u + at^2$
 (c) $v = u + 2at$ (d) $v = ut + 1/2at^2$
4. If an object moves in a circular path at a constant speed, what can be said about its acceleration?
 (a) It is directed toward the center of the circle.
 (b) It is directed away from the center of the circle.
 (c) It is zero.
 (d) It is directed tangentially to the circle.
5. If $a \cdot b = 0$ when $a \neq 0$, $b \neq 0$, then the vectors are .
 (a) Parallel. (b) Opposite.
 (c) Perpendicular. (d) Zero.
6. When an object is at rest, what is the value of its velocity?
 (a) Zero (b) Positive
 (c) Negative (d) Cannot be determined from the information given.
7. If a car accelerates uniformly from 10 m/s to 30 m/s in 5 seconds, what is its acceleration?
 (a) 4 m/s^2 (b) 6 m/s^2 (c) 8 m/s^2 (d) 10 m/s^2
8. If A and B are two vector
 (a) $A \cdot B = B \cdot A$ (b) $A \cdot B = -B \cdot A$
 (c) $A \cdot B = -A \cdot B$ (d) $B \cdot A = -B \cdot A$
9. Two bodies launched with different angle from a same point and they had same range. The angle will be:
 (a) 45° and 30° (b) 30° and 70°
 (c) 20° and 70° (d) 100° and 80°
10. The motion of a curved path when one component velocity is constant and other is variable called
 (a) Variable motion. (b) Projectile motion.
 (c) Vibratory motion. (d) Circulatory motion.

11. If a projectile is fired at angle of 45° with the velocity of 100 m/s, it hits the target. It will have double the range if its velocity is (a) 141.4 m/s. (c) 1173.2 m/s.	(b) 200 m/s. (d) 400 m/s.	
12. The range of a projectile becomes half of its maximum value if the angle of projection is (a) 15° (c) 60°	(b) 45° (d) 90°	a
13. Angular velocity is also known as angular (a) Rotation. (c) Frequency.	(b) Speed. (d) Acceleration.	c
14. A car is moving with a speed of 20 m/s. It increases its speed to 40 m/s in 5 seconds. What is the magnitude of its acceleration? (a) 12 m/s^2 . (c) 16 m/s^2 .	(b) 8 m/s^2 . (d) 4 m/s^2 .	d
15. A body goes from 2 meter to 8-meter mark and back to 2 meter mark in 3 sec, its average speed is (a) 2 m/s (c) 4m/s	(b) 6m/s (d) Zero	c
16. Two forces of the same magnitude F make an angle of 180° with each other their resultant is (a) zero (c) $\sqrt{2} F$.	(b) F (d) 2F	a
17. The dot product of unit vectors i and k is (a) Zero (c) j	(b) 1 (d) -1	a
18. Due to presence of air resistance the total time of flight of a projectile: (a) Remains the same (c) Becomes zero.	(b) Decreases. (d) Increases.	d
19. If a vector quantity is divided by its magnitude, the vector obtained is called (a) Unit vector. (c) Null vector.	(b) Position Vector. (d) Free vector.	a
20. What does a horizontal line on a velocity-time graph represent? (a) Constant acceleration (c) Negative acceleration	(b) Zero velocity (d) Changing velocity	b
21. On a velocity-time graph, the area under the curve represents: (a) Displacement (c) Time	(b) Acceleration (d) Speed	a
22. A diagonal line on a velocity-time graph indicates: (a) Constant velocity (c) Decreasing velocity	(b) Increasing velocity (d) Zero velocity	a
23. If a velocity-time graph is a straight horizontal line, what can be said about the object's motion?		

(a) The object is at rest. (b) The object is accelerating. (c) The object is moving with constant speed. (d) The object is changing direction continuously	a
24. On a displacement-time graph, a curve that is steeper indicates: (a) Slower motion (b) Faster motion (c) Negative displacement (d) No motion	b
25. What does the slope of a displacement-time graph represent? (a) Speed (b) Acceleration (c) Distance (d) Velocity	d
26. When $\vec{A} - \vec{B} = \vec{A} + \vec{B}$ the angle between the vectors $\vec{A}$ and $\vec{B}$ is (a) Zero. (b) 45°. (c) 90°. (d) 180°.	c
27. On a displacement-time graph, the area under the curve represents: (a) Speed (b) Velocity (c) Acceleration (d) Distance travelled	d
28. Two perpendicular vectors having the magnitudes of 4 units and 3 units are added. The resultant of the magnitude is (a) 7 Units (b) 12 Units (c) 25 Units (d) 5 Units	d
29. If $\vec{A} \cdot \vec{B} = \vec{A} \times \vec{B}$, then the angle between A and B is (a) 90° (b) 45°. (c) 120°. (d) 180°.	a
30. $\mathbf{j} \times \mathbf{j}$ is equal to (a) $\mathbf{j}^2$ (b) $\mathbf{j}$. (c) Zero (d) one	c

VECTORS:

Commutative property of Vector Addition

$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

Associative Property of Vector Addition

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$$

Resultant by Cosine Law

$$R = \sqrt{A^2 + B^2 - 2AB \cos \angle OAB}$$

Resultant by Sine law

$$\frac{A}{\sin \alpha} = \frac{B}{\sin \beta} = \frac{R}{\sin \theta}$$

Unit Vector

$$\hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

Rectangular Components of vector

- (i) horizontal component: $F_x = F \cos \theta$
 (ii) vertical component $F_y = F \sin \theta$

Direction of Vector

$$\theta = \tan^{-1} \left(\frac{F_y}{F_x} \right)$$

DOT PRODUCT:

- (i) $\vec{A} \cdot \vec{B} = AB \cos \theta$
 (ii) $\vec{A} \cdot \vec{B} = A (\text{Projection of } \vec{B} \text{ on } \vec{A})$

$$\vec{A} \cdot \vec{B} = AB_A \quad \forall B_A = B \cos \theta$$

Properties of Dot Product

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$$

$$\hat{i} \cdot \hat{j} = \hat{k} \cdot \hat{j} = \hat{k} \cdot \hat{i} = 0$$

CROSS PRODUCT:

i) $A \times B = AB \sin \theta$

ii) $A \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

Properties of Cross Product

$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 0$

$\hat{i} \cdot \hat{j} = \hat{k}, \hat{j} \cdot \hat{i} = -\hat{k}$

$\hat{j} \cdot \hat{k} = \hat{i}, \hat{k} \cdot \hat{j} = -\hat{i}$

$\hat{k} \cdot \hat{i} = \hat{j}, \hat{i} \cdot \hat{k} = -\hat{j}$

MOTION:

Velocity:

$\vec{v} = \frac{\Delta \vec{s}}{\Delta t}$

$V_{\text{ins}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t}$

Acceleration:

$\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$

$\vec{a}_{\text{int}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t}$

Equation of Motions

i) $V_f = V_i + at$

ii) $S = vit + \frac{1}{2}at^2$

ii) $2as = v_f^2 - v_i^2$

PROJECTILE MOTION:

Time of flight : $t = \frac{v_0 \sin \theta}{g}$

Total time of flight $T = \frac{2v_0 \sin \theta}{g}$

Height of Projectile: $H = \frac{v_0^2 \sin^2 \theta}{2g}$

Range of projectile $R = v_0^2 \sin 2\theta / g$

Max. Range

or

$\therefore R_{\text{max}} = 4H$

$R_{\text{max}} = \frac{v_0^2}{g}$

Coefficient of Friction

$\mu = \frac{F}{R}$

Angle of Friction

$\tan \alpha = \mu$